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# Residential Segregation and Racial Cancer Disparities: A Systematic Review

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**Abstract**
*Background* This paper provides the first review of empirical studies of segregation and black-white cancer disparities.
*Methods* We searched all years of PubMed (through May 2016) using these terms: racial segregation, residential segregation, neighborhood racial composition (first terms) and (second terms) cancer incidence, mortality, survival, stage at diagnosis, screening. The 17 (of 668) articles that measured both segregation and a cancer outcome were retained.
*Results* Segregation contributed significantly to cancer and to racial cancer disparities in 70% of analyses, even after controlling for socioeconomic status and health insurance. Residing in segregated African-American areas was associated with higher odds of later-stage diagnosis of breast and lung cancers, higher mortality rates and lower survival rates from breast and lung cancers, and higher cumulative cancer risks associated with exposure to ambient air toxics. There were no studies of many types of cancer (e.g., cervical). Studies differed in their measure of segregation, and 40% used an invalid measure. Possible mediators of the segregation effect usually were not tested.
*Conclusions* Empirical analysis of segregation and racial cancer disparities is a recent area of research. The literature is limited to 17 studies that focused primarily on breast cancer. Studies differed in their measure of segregation, yet segregation nonetheless contributed to cancer and to racial cancer disparities in 70% of analyses. This suggests the need for further research that uses valid measures of segregation, examines a variety of types of cancers, and explores the variables that may mediate the segregation effect.

**Keywords** Residential segregation · Race · Cancer · African-Americans · Blacks · Whites · Cancer disparities

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## Introduction

**Rationale** There is growing interest in the role of neighborhoods in cancer and cancer disparities [1–3]. This recent focus reflects mounting evidence that cancers are unequally distributed across geographic areas that differ in socioeconomic status (SES), racial-ethnic composition, and rurality [1–3]. Ascertaining the role of neighborhoods in cancer can identify communities in need of healthcare services and might suggest new strategies for reducing cancer disparities [1]. Because African-Americans (AAs) and whites generally reside in separate and unequal neighborhoods [1, 4], neighborhood variables may be particularly relevant to racial cancer disparities [1, 5]. Despite recent declines in cancer incidence and mortality, significant racial disparities in both remain, and AA cancer mortality rates exceed those of all racial-ethnic groups [5, 6]. This review examines empirical studies of racial residential segregation and cancer among AAs and whites for the first

time. Although there are many studies [e.g., 7–12] and a few comprehensive reviews [2, 9, 10] of the role of neighborhood-SES in cancer and cancer disparities, there are no similar reviews of segregation [1]. Hence, we first describe segregation, its measurement, and its relationship to health disparities, and then we review empirical studies of segregation and cancer.

**Objectives** The objective of this review was to provide answers to these five questions: Does racial residential segregation contribute to cancer incidence, mortality, survival, risk, screening, and/or stage at diagnosis? What measures of segregation were used? What types of cancers were examined? Did studies of cancer patients focus on a single state or on multiple states? What are the moderators/mediators of segregation's role in cancer?

## Racial Residential Segregation

Residential segregation refers to the geographic separation of racial-ethnic minorities (AAs here) from whites in residential areas [4, 13, 14]. Segregation can be measured at any geographic area level [4] including census tracts (CTs), zip codes, counties, states, and metropolitan statistical areas (MSA). Segregation effects on health usually are stronger when segregation is measured in small areas (e.g., comparing census tracts) than in large ones (e.g., comparing states or counties) because large areas include both low- and highly- segregated smaller areas within them [4, 13, 14]. Irrespective of geographic area level, six measures of segregation are widely regarded as valid measures. These are: (1) *dissimilarity* (the uneven distribution of AAs and whites in residential areas); (2) *isolation* (the average probability that AAs will encounter only other AAs in their residential area); (3) *concentration* (the population density of AA neighborhoods); (4) *clustering* (the extent to which AA neighborhoods are surrounded by other AA neighborhoods); (5) *centralization* (the degree to which AA neighborhoods are located in urban centers); and (6) *hypersegregation*, the simultaneous occurrence of all of these [4, 15–17]. The isolation index is viewed as psychometrically superior to the other measures and hence is the most commonly used of the valid measures [13, 16, 18–20]. Isolation scores range from 0 (completely integrated) to 1 (totally segregated) with ≥0.6 regarded as high and ≥0.7 regarded as very high segregation [13–20]. The dissimilarity or segregation index (SI, calculated by the US Census Bureau and available in census datasets) also is often used and has the same range and segregation thresholds as the isolation index. Crude measures of segregation that have dubious validity sometimes are used as well [4, 13, 16]. The percent of AAs in an area (with ≥25% defined as high segregation) is one such measure. Its validity is suspect because an area can consist of up to 50% AAs yet not be highly segregated (i.e., isolation index is <0.60) if AAs and whites live near each other and encounter each other often [4, 13, 16, 19] Table 1 provides evidence of this, based on our calculations using 2013 census data on census tracts (CTs) in Pitt County, NC. As will be shown, many studies reviewed used percent of AAs as their segregation measure.

**Segregation of AAs vs. Other Minorities** Dissimilarity (SI) scores often are multiplied by 100 to range from 0 (fully integrated) to 100 (a totally segregated MSA in which AAs and whites reside in separate neighborhoods). SI data are interpreted as the percentage of AAs who would have to move to integrate the city or MSA [4, 13–19]. SI data reveal that AA segregation continues to be mostly high (SI ≥60) or very high (SI ≥70). For example, 2010 AA SIs are Milwaukee 81.5, Detroit 75.3, and Chicago 76.4—i.e., 75 to 81% of AAs would have to move to integrate the aforementioned cities [4]. AAs also are more segregated than other racial-ethnic minority groups. For example, 2010 SIs for Latinos are Milwaukee 57.0, Detroit 43.3, and Chicago 65.3 [4]; for Asians, they are Milwaukee 40.0, Detroit 50.6, and Chicago 44.9 [4]. Consequently, segregation might play a greater role in the health of AAs than of other racial-ethnic groups. The high level of segregation of AAs does not reflect their preferences [4, 21–23] but instead is the result of discriminatory housing practices [4, 24–28]. These practices hold across AA income and education, and hence also affect high-SES AAs, e.g., those who attempt to purchase homes in affluent, white suburbs [21, 29–33]. Thus, residential racial segregation does not mirror residential income segregation and is not an artifact of racial differences in SES [21–33]. Indeed, if the US population were distributed into residential areas based solely on income, the USA would be highly integrated [13–20]. For example, based on income alone, the 1990 and 2000 AA SI for St. Louis would be 11 instead of >70 [32]. Importantly then, residing in segregated AA areas may in part explain why AA health disparities often remain after controlling for SES [1, 4, 18].

**Table 1** Percent African-American (AA) and isolation index for selected census tracts (CTs) in Pitt County, NC

| CT | N | AAs (%) | Whites (%) | AA isolation index |
|---|---|---|---|---|
| 1301 | 3883 | 1307 (33.7) | 2382 (61.3) | 0.45 |
| 1500 | 3315 | 1159 (35.0) | 1950 (58.8) | 0.37 |
| 0603 | 9570 | 3694 (38.6) | 5068 (53.1) | 0.42 |
| 1401 | 4801 | 1982 (41.3) | 2522 (52.5) | 0.50 |
| 1800 | 6120 | 2817 (46.0) | 2975 (48.6) | 0.57 |
| 2001 | 6287 | 3143 (50.0) | 1825 (29.0) | 0.54 |

Calculated by the authors from NC Census data at www.census.gov

**Segregation and AA Health Disparities** AA segregation has been shown to contribute to AA disparities in infectious disease [34], asthma [35, 36], adult mortality [37], hypertension [38], cardiovascular disease [39, 40], adverse birth outcomes [41, 42], obesity [43–45], cigarette smoking [1, 4], and other outcomes [1, 4, 45–48]. The built environments of such neighborhoods are widely regarded as the mediator of the segregation effect on health. For example, AA neighborhoods, compared to SES-matched white neighborhoods, have significantly higher environmental exposures, higher access to tobacco and exposure to tobacco advertising, and significantly lower access to healthy food, physical activity resources, and quality healthcare [4, 18, 34–48].

## Method

**Literature Search** We searched all years of PubMed (through May 2016) using these terms: racial segregation, residential segregation, neighborhood racial composition (first terms) and (second terms) cancer incidence, cancer mortality, cancer survival, cancer stage at diagnosis, cancer screening, cancer risk. Additional searches used the first terms with specific cancers as the second terms (e.g., prostate cancer incidence, prostate cancer mortality, breast cancer incidence, etc.). The combined searches returned 668 articles. Their abstracts were examined by the authors.

**Exclusion Criteria** Articles that were theoretical, commentaries, dissertations, did not measure segregation and did not measure a cancer outcome were excluded. Only studies that measured both residential segregation (using any measure of segregation) and a cancer outcome (incidence, mortality, survival, stage at diagnosis, screening, risk) were retained. Using these criteria, 651 studies were excluded, and the 17 empirical analyses of segregation and a cancer outcome were retained.

**Studies Selected** The 17 studies are described in Table 2, where their year of publication, sample sizes, geographic location, measure of segregation, database, and major findings also are summarized. As shown in Table 2, the studies were published 2006–2016, with most (15 of 17) published 2010–2016; no earlier studies were found. All 17 studies were secondary data analyses of a well-known, reliable and valid, existing dataset, as detailed below and in Table 2. Datasets used in the 17 studies were from 1992 to 2009.

## Results

**Quality of Studies and Risk of Bias** Of the 17 studies summarized in Table 2, 13 focused on diagnosed cancer cases. The 13 studies of cancer patients [59–71] were all secondary-data analyses of a cancer-registry dataset for the years shown in Table 2. A cancer registry is a standardized collection of data about cancer. Such data are collected in states by highly trained Cancer Registrars who transfer information from patients' medical records to the cancer registry using the variables, codes, and software specified in and provided at no cost by the Centers for Disease Control (CDC's) National Program of Cancer Registries (NPCR; see http://www.cdc.gov/cancer/npcr). For all cases of diagnosed cancer, state cancer registries include demographic information, diagnostic data, treatment details, and follow-up information on mortality and survival. Thus, all 13 cancer-registry studies contained a large sample, included the same demographic variables (e.g., gender, age, health insurance) and controlled for those in analyses, and included and controlled for the same cancer variables (e.g., tumor stage, grade, histology); some studies also controlled for healthcare (access, quality). The design, quality, and variables in the 13 cancer-registry studies were largely identical; this holds for studies that used cancer-registry data from a single state, and for those that used national or multi-state cancer-registry data. Each year, state cancer registries electronically submit their data to the CDC's NPCR and to the National Cancer Institute's Surveillance, Epidemiology, and End Results (SEER) Program, thereby yielding national and multi-state cancer-registry datasets complied from single-state datasets. The multi-state cancer-registry studies reviewed here used national SEER data or used SEER data on the commonly analyzed set of selected states (i.e., data from the cancer registries of California, Connecticut, Hawaii, Iowa, Kentucky, Louisiana, New Jersey, New Mexico, Utah, rural Georgia, and metropolitan Detroit, Atlanta, and Seattle). Given that the variables, measures, and designs of the 13 cancer registry studies were essentially identical, we do not compare and contrast the studies on those study features. This also means that study design, sample, and quality do not explain differences between the studies in their findings. The cancer registry studies differed only in their measure of segregation and in use of a single-state or multiple-state registry, and we compare and synthesize the studies on those variables. Any risk of bias in the 13 cancer-registry studies is a function of limitations of cancer-registry data and limitations of the census data that were joined to those to analyze segregation, discussed later here.

The remaining 4 of 17 studies (Table 2) examined cumulative cancer risks associated with exposure to ambient air toxics [72–75]. All 4 studies were secondary analyses of Environmental Protection Agency (EPA) data on ambient air toxics. Each study used standardized data and measures from the EPA's National Air Toxics Assessment (NATA) Project (and similar projects) on air toxics throughout the USA [62], and each calculated cancer risk using standardized, well-accepted EPA measures (e.g., inhalation unit risk, micrograms

**Table 2** Empirical studies of racial segregation and cancer among African-Americans (AAs) and whites

| # | Reference | Dataset | Sample | Cancer (CA) | Segregation measure | Results |
|---|---|---|---|---|---|---|
| Lung cancer | | | | | | |
| 1 | Hayanga et al. 2013 [49] | 2003–2007 nationwide SEER | 3452 counties | Lung CA mortality | Dissimilarity at the county level with segregation levels of low <40, moderate = 40–59, and high ≥60 | Adjusted mortality rates (per 100,000) for African-Americans (AA) and whites (AA = 52.4, whites = 53.0) were identical in low-segregated counties, but were significantly higher for AAs in moderately (AAs = 57.0, whites = 51.1) and high-segregated counties (AAs = 62.9, whites = 50.0). There was a statistically significant 0.5% increase in AA mortality rates and a similarly significant 0.2% decrease in white mortality rates with increasing segregation. These effects remained after controlling for county smoking, SES, , and cancer variables (e.g., tumor stage, grade, histology). Other possible mediators of the segregation effect were not tested. |
| 2 | Haas et al. 2008 [50] | 1992–2002 SEER | 81,508 | Lung CA stage at diagnosis | Isolation at the census tract (CT) level | Whites were significantly more likely to be diagnosed at an early stage than AAs. Early stage diagnosis generally was not related to CT-SES or CT-segregation for either race. However, AAs in high-segregated/low-income CTs were significantly less likely than whites and AAs in other CTs to be diagnosed at an early stage. Possible moderates of the segregation effect were not tested. |
| 3 | Johnson et al. 2016 [51] | 2000–2009 GA cancer registry | 6867 whites 1455 AAs | Lung CA surgery and survival rates | Isolation | Segregation was the strongest predictor of surgery among AAs (i.e., stronger than area-SES and individual-level variables). AAs in high-segregated areas were 63% less likely to receive surgery, and AAs in high-segregated/low-income areas were 31% more likely die even when controlling for surgery and cancer variables. Other possible moderates of the segregation effect were not tested. |
| Prostate cancer | | | | | | |
| 4 | DeChello et al. 2006 [52] | 1994–1998 CT & MA cancer registry | 29,040 whites 1647 AAs | Prostate CA incidence | % AAs | No relationship between segregation and prostate cancer incidence |
| 2a | Haas et al. 2008 [50] | 1992–2002 SEER | 151,142 | Prostate CA stage at diagnosis | Isolation | No relationship between segregation and prostate cancer stage at diagnosis |
| Breast cancer | | | | | | |
| 5 | Russell et al. 2012 [53] | 1999–2003 GA cancer registry | 17,289 whites 4799 AAs | Breast CA mortality | CT and MSA % AAs | As MSA segregation increased, breast cancer mortality rates increased significantly among AA women but not among white women. Possible mediators of this effect were not examined. |
| 6 | Pruitt et al. 2015 [54] | 1995–2009 TX cancer registry | 82,089 whites 11,253 AAs | Breast CA mortality | CT and MSA % AAs | As segregation increased, breast cancer mortality increased among AA and white women. Cancer variables and area-SES were controlled and did not mediate the segregation effect. Other possible mediators were not tested. |
| 7 | Russell et al. 2011 [55] | 1999–2003 GA cancer registry | 11,775 whites 3481 AAs | Breast CA mortality and care | CT % AAs | AA women were 1.5 times more likely to die from breast cancer than white women. Residing in segregated AA neighborhoods was associated with significant increased breast cancer mortality irrespective of race; access to cancer care did not explain these outcomes. |
| 8 | Haas et al. 2008 [56] | 1999–2002 SEER | 43,287 whites 2875 AAs | Breast CA mortality and care | CT level isolation | AA women had significantly higher mortality rates and lower survival rates than white women. AA women were significantly less likely than white women to receive adequate breast cancer care (defined as either mastectomy or breast conserving surgery with radiation therapy within 6 months of diagnosis, and assessment of estrogen receptor status at the time of surgery). Both AA and white women in AA-segregated CTs were less likely to receive adequate care than those in non-segregated CTs. Segregation mediated racial disparities in cancer care, but did not explain racial disparities in mortality and survival. |
| 2b | Haas et al. 2008 [50] | 1992–2008 SEER | 86,723 | Breast CA stage at diagnosis | CT level isolation | White women were significantly more likely than AA women to be diagnosed at an early stage irrespective of CT-SES and segregation. Segregation did not contribute to disparities in stage at diagnosis; cancer variables were controlled. |
| 9 | Warner et al. 2010 [57] | 1996–2004 CA cancer registry | 113,979 whites 10,030 AAs | Breast CA stage at diagnosis | % AAs and any two valid measures | AA women were significantly more likely than white women to be diagnosed with later-stage breast cancer; segregation did not mediate or explain this racial disparity. |
| 10 | Dai 2010 [58] | 1998–2002 MI cancer registry | 12,413 | Breast CA stage at diagnosis | Isolation at the zip code level | Segregation, access to mammography, and access to primary care were each independently and significantly associated with late-stage diagnosis of breast cancer among AA women. |

**Table 2** (continued)

| # | Reference | Dataset | Sample | Cancer (CA) | Segregation measure | Results |
|---|---|---|---|---|---|---|
| 11 | Mobley et al. 2008 [59] | 2002–2003 SEER data from 11 states | 224,585 | Breast CA screening | Isolation at the CT and MSA level | AA women in high-segregated areas in three states (CA, IA, WA) had lower odds of mammography than their low-segregated cohorts, whereas in one state (KY), AA women in high-segregated areas had higher odds of mammography, and in four states (NM, LA, CT, or NJ), segregation did not contribute to mammography. Mediators of state-level variation in segregation effects were not tested. |
| Other cancers | | | | | | |
| 2c | Haas et al. 2008 [50] | 1992–2008 SEER | 91,497 | Colorectal CA stage at diagnosis | CT level isolation | Segregation did not contribute to racial disparities in stage at diagnosis of colorectal cancer. |
| 12 | Hao et al. 2011 [60] | 2000–2004 GA cancer registry | 3575 whites 1173 AAs | Chemo for colorectal CA | CT level % AAs | Rural vs. urban vs. suburban residence and race contributed to receiving chemotherapy for colorectal cancer, and segregation did not contribute to treatment disparities. |
| 13 | Hao et al. 2011 [61] | 2000–2003 11 state cancer registries | 4780 white, 415 AA survivors of 10 types of CA | CA-related quality of life | County level isolation | Health-related quality of life (HRQL) was significantly poorer among AA than white cancer survivors. When segregation was controlled, this racial disparity disappeared: HRQL was significantly poorer among survivors who resided in AA-segregated areas irrespective of race, and area-SES played no role. Cancer registries used were from Alabama, Connecticut, Iowa, Idaho, Minnesota, New Jersey, Ohio, Pennsylvania, Rhode Island, South Carolina, and Wyoming. The 10 cancers examined were female breast, prostate, urinary bladder, uterine, skin melanoma, colorectal, kidney, non-Hodgkin's lymphoma, ovarian, and lung. |
| Cancer risk associated with ambient air toxics | | | | | | |
| 14 | Morello-Frosch and Jesdale 2006 [62] | 2002, 2004 Nationwide EPA NATA (National Air Toxics Assessment) data | 45,710 CTs | CA risk | CT dissimilarity | AA cumulative cancer risk increased with increasing segregation |
| 15 | Rice et al. 2014 [63] | 2005 EPA NATA data for SC | 105 CTs | CA risk | CT isolation | AA cumulative cancer risk increased with increasing segregation |
| 16 | Apelberg et al. 2005 [64] | 2002, 2004 EPA NATA data for MD | 1210 CTs | CA risk | CT % AAs | AA cumulative cancer risk increased with increasing segregation |
| 17 | Linder et al. 2008 [65] | 1999 EPA NATA data for TX | 62 CTs | CA risk | CT % AAs | AA cumulative cancer risk increased with increasing segregation |

The 17 studies above entailed 20 analyses because the Haas study [50] conducted three additional analyses, shown above as 2a, 2b, and 2c

per cubic meter of specific carcinogens). Samples were large, and area-SES, area-poverty, and other potentially confounding area variables were controlled. The cancer-risk studies were similar in approach, quality, variables, and dataset and hence we do not compare the studies on those study-features. The four studies differed only in their sample sizes and measure of segregation. Any risks of bias in these studies reflect possible bias and limitations of the census dataset that was joined to the EPA data, discussed later here.

**Lung Cancer** AAs have the highest lung cancer incidence and mortality rates [5, 6, 76, 77]. As shown in Table 2, there are only three studies of segregation and lung cancer [49–51]. The first found that racial disparities in mortality rates increased with increasing segregation. The second [50] found that AAs who resided in high-segregated/low-income census tracts were less likely than whites and AAs residing elsewhere to be diagnosed at an early stage. The third study [51] found that AAs in high-segregated areas were significantly less likely than whites to receive surgery and were more likely than whites to die even when controlling for surgery. Thus, all three studies of lung cancer disparities found segregation to be a significant contributor.

**Prostate Cancer** Prostate cancer incidence rates are 70% higher in AA than in white men [5]. AA men also have higher prostate cancer mortality rates than all racial-ethnic groups, and their mortality rate is 2.4 times higher than that of white men [5]. Several studies [e.g., 78–80] have examined the role of area-level factors in prostate cancer but did not examine segregation. As shown in Table 2, there are only two studies of segregation and prostate cancer. The first [52] found no relationship between segregation and prostate cancer incidence, and the second [50] found no relationship between segregation and prostate cancer stage at diagnosis, for white and AA men. Thus, in both studies of prostate cancer, segregation played no role.

**Breast Cancer Mortality** Breast cancer incidence rates are 3% lower in AA than white women, with two exceptions: Incidence rates are higher among AA women in seven states (Alabama, Kentucky, Louisiana, Mississippi, Missouri, Oklahoma, and Tennessee) for unknown reasons [5] and are higher among AA than among white women under age 45 [5, 6]. Incidence among white women stabilized 2004–2012 but increased among AA women by 0.5% each year [5, 6]. Consequently, racial incidence rates converged in 2012 [5]. These trends have been attributed to increasing racial disparities in (significantly higher AA rates of) obesity, a major risk factor for breast cancer [5, 6]. Despite generally lower incidence rates, breast cancer mortality remains 42% higher among AA than among white women [5]. This has been explained as a function of AA women's later stage at diagnosis, higher prevalence of comorbidities, more aggressive (e.g., triple negative) tumors, higher race-related and financial stress with associated chronic inflammation, and lower access to quality cancer care [5, 81, 82].

As shown in Table 2, there were no studies of breast cancer incidence, but there were four studies of segregation and breast cancer mortality [53–56]. Three found that increasing segregation was associated with increasing mortality for AA women [53] or for both AA and white women [54, 55], and one found no effect. Thus, three of four studies of racial disparities in breast cancer mortality found segregation to be a significant contributor.

**Breast Cancer Stage at Diagnosis and Screening** Three studies (Table 2) analyzed segregation and racial disparities in breast cancer stage at diagnosis [50, 57, 58]. Two [50, 57] found that segregation did not contribute to stage at diagnosis, whereas one [58] found that segregation played a significant role in late-stage diagnosis among AA women. Thus, two of three studies of segregation and racial disparities in breast cancer stage at diagnosis found that segregation did not play a role. In addition, the sole study of segregation and breast cancer screening [59] found that *segregation contributed to racial disparities in mammography use in some states but not in others*.

**Other Cancers** Table 2 highlights the three studies of segregation and other types of cancer. Two studies [50, 60] examined colorectal cancer and found that segregation did not contribute to disparities in stage at diagnosis [50] or to use of chemotherapy [60]. A third study [61] found that segregation predicted poor health-related quality of life among both black and white survivors of 10 types of cancers (combined); initial racial disparities in cancer survivors' quality of life disappeared when segregation was statistically controlled.

**Cancer Risk** Four studies examined the role of segregation in racial disparities in cancer risks associated with exposure to air toxics [62–65]. In a nationwide US study [62], area-SES, segregation (dissimilarity scores), and inhalation unit risk were analyzed in 309 MSAs that encompassed 45,710 CTs and included >79% of the US white and 85% of the US AA population. High-segregated AAs had significantly higher exposures to air toxics and higher cumulative cancer risks than whites and low-segregated AAs; this effect increased with increasing AA segregation and was independent of area-SES. Three smaller, similar studies in South Carolina [63], Maryland [64], and Texas [65] found identical results (see Table 2). Hence, all four cancer-risk studies found effects for segregation.

## Synthesis

**Overall Synthesis** The 17 empirical studies of racial segregation and cancer entailed 20 separate analyses. This is because the Haas study [50] conducted four analyses (one for each of four different cancers, see Table 2). Of the 20 analyses, 6 found no effect for segregation; 9 found that residing in segregated AA areas contributed to cancer and cancer disparities among AAs; 4 found that residing in segregated AA areas contributed to cancer among AAs and whites alike; and 1 found segregation effects for AAs in some but not in other US states. Thus, in 70% of (14 of 20) analyses, segregation contributed significantly to cancer and racial cancer disparities.

**Mediators/Moderates Examined** The mechanism (i.e., potential mediators/moderators) of the segregation effect on cancer was empirically tested in only 3 of the 13 (23% of) cancer-registry studies. Each of those studies examined access to and quality of cancer care as possible mediators and found that such variables did not mediate or explain the segregation effect on racial disparities in breast [55, 56] and lung [51] cancer. Most (10 of 13 or 77% of) cancer registry studies did not test any of myriad potential mediators.

**Segregation Measures Used** Twelve of 20 (60% of) analyses used valid segregation measures (dissimilarity or isolation) and found effects in 75% of analyses. Eight of 20 (40% of) analyses used percent AAs (an invalid measure) as the measure of segregation and found effects in 62% of analyses.

**Single vs. Multiple-State Cancer Studies** Eleven analyses focused on cancer-registry data for a specific US state, and a significant segregation effect was found in eight (73%) of those. Seven analyses used the combined cancer-registry data from 9 to 11 states (i.e., SEER data), and only 2 (29%) found a segregation effect.

## Discussion

**Summary of Evidence** We summarize the evidence in terms of the five objectives/questions of this review.

**Objective/Question 1** *Does racial residential segregation contribute to cancer incidence, mortality, survival, risk, screening, and/or stage at diagnosis?* The answer is that segregation contributed significantly to cancer and to racial cancer disparities in 70% of the analyses reviewed, even after controlling for health insurance and for area-SES. Residing in segregated AA areas was associated with a higher probability of late-stage diagnosis of breast and lung cancers, higher mortality rates and lower survival rates from breast and lung cancers, and higher cumulative cancer risks associated with exposure to ambient air toxics. Although there were only 17 studies and the majority were recent (published 2010–2016), the studies strongly suggest that racial residential segregation contributes to cancer and to racial cancer disparities.

**Objective/Question 2** *What measures of segregation were used?* We found that 60% of analyses used the isolation index or the dissimilarity index (both valid measures of segregation), and that significant segregation effects were found in 75% of analyses that used these measures. In addition, however, 40% of analyses used percentage of AAs in an area as the measure of segregation despite evidence (and the examples shown in Table 1) that its validity is dubious at best [4, 13, 16, 19].Significant segregation effects were found in 62% of such analyses. Different, potentially stronger and more consistent relationships between segregation and cancer might have been found if all analyses had used valid measures of segregation. Indeed, only 50% of analyses used the most robust segregation measure, the isolation index. Moreover, the four other, well-known and valid measures of segregation detailed earlier here (concentration, clustering, centralization, hypersegregation) have not been used in studies of segregation and cancer. Use of these might highlight complex, more nuanced segregation effects. The measures of segregation used constitute a major limitation of this area of research.

**Objective/Question 3** *What types of cancers have been examined?* We found that 10% of analyses focused on prostate cancer (2 of 20), 10% on colorectal cancer, 15% on lung cancer (3 of 20), 20% on environmental cancer risks (4 of 20), 5% on 10 combined cancers that could not be differentiated (1 of 20), and 40% focused on female breast cancer (8 of 20). More studies and analyses focused on female breast cancer than on lung, prostate, and colorectal cancers combined, even though racial disparities in incidence and (especially) mortality rates from lung, prostate, and colorectal cancers are higher than those for breast cancer (as detailed earlier here). Importantly, there were no studies of prostate cancer mortality; no studies of colorectal cancer screening, incidence or mortality; and only one study of incidence rates of any type of cancer. Moreover, despite well-known, persistent racial disparities, there were no studies of many cancers such as cervical, liver, and ovarian [1, 5, 6]. The small number of types of cancers examined, the failure to devote equal attention to each of the four most common cancers (lung, prostate, colorectal, female breast), and the paucity of studies of cancer incidence are limitations of this area of research.

**Objective/Question 4** *Did the studies of cancer patients focus on a single state or on multiple states?* We found that 73% of the studies that used cancer-registry data from a specific US

state found significant effects for segregation, whereas only 29% of those that used multi-state SEER data on combined states found a segregation effect. We interpret this difference as follows: Racial residential segregation originated in efforts to restore white supremacy in the wake of AA emancipation from slavery [15, 73]. Segregation and its health-effects vary with the specific methods used originally and subsequently by each state to perpetuate that supremacy [15, 73], and there are well-known differences in the methods used by the former (southern) slave states versus the northern states to which emancipated AAs later migrated [13, 22, 23, 73, 74]. Differences between states in their historical and current methods and levels of segregation probably explain why segregation effects were found in most of the state-specific analyses but rarely were found in the multi-state analyses. Given the state-level variation in segregation levels and methods, it may be beneficial for future analyses to use cancer-registry data for individual states. The findings of the Mobley study [59] support this view: Mobley found that segregation contributed positively, negatively, or not at all to racial disparities in mammography depending on US state. Future studies that analyze cancer data by state also may find that the most robust segregation measure (each representing a slightly different segregation method) varies by state. This possibility never has been examined. Thus, the use of multi-state SEER data is a major limitation of this area of research; such data were used in most (8 of 13 or 62%) of the analyses of cancer-registries.

**Objective/Question 5** *What are the moderators/mediators of segregation's role in cancer?* As noted earlier, only 3 of the 13 cancer-registry analyses examined possible mediators, and the mediators in all three were access to and quality of cancer care. These variables did not mediate or explain the segregation effect on racial disparities in breast [55, 56] and lung [51] cancer. Most (10 of 13 or 77% of) cancer registry studies did not test any of numerous, potentially robust mediators. Consequently, how and why segregation contributes to cancer and racial cancer disparities remains unknown. The failure to empirically examine mediators of segregation's role in cancer is a major limitation of the segregation-cancer-registry studies, and probably reflects the absence of health-behavior variables in most cancer registries.

Specifically, three (of myriad potential) mediators that have not been examined in the cancer-registry studies are: (1) physical activity, which is known to contribute to incidence, mortality, and survival rates of a variety of cancers [61, 75]. Our prior studies [1, 45] demonstrated that the physical activity level of AAs who reside in segregated areas is significantly lower than that of those who reside elsewhere, no doubt because of the dearth of sidewalks, street lighting, and recreational facilities in segregated, AA neighborhoods [1, 45, 48]. (2) Obesity contributes to incidence, mortality, and survival of numerous cancers through a variety of biological mechanisms [5, 6, 75]. We have demonstrated that obesity rates are significantly higher among segregated than among non-segregated AAs [1, 43, 45] as a function of the low access to healthy food and to recreational resources in segregated-black areas [1, 46, 48]. (3) Tobacco use not only causes lung cancer, but also contributes causally to cancers of the cervix, colon, rectum, pancreas, kidney, bladder, liver, stomach, and other areas, and tobacco use is responsible for one out of every three cancer deaths in the USA [76, 77]. We have found that smoking prevalence rates among segregated AAs significantly exceed those of their non-segregated counterparts [1, 78]. Thus, these three health behaviors/variables may mediate the segregation effect on cancer among AAs who reside in segregated-AA areas.

Data on physical activity, obesity, and tobacco use are provided in patient medical records, but usually are not included in the cancer registries based on those records; hence, these variables have not been examined. Cancer registries include demographic, diagnostic, and treatment information but rarely also include information on the many health behaviors that are known to be relevant to cancer. The narrow range of variables in the majority of cancer registries is a major limitation of research on segregation and cancer, and obstructs investigation of the mediators of the segregation effect. Knowledge of the role of health behaviors in cancer stems (not from cancer-registry data but) primarily from other datasets and studies.

**Limitations of the Studies** Limitations noted above are (1) frequent use of invalid measures of segregation and failure to use the array of valid measures; (2) the small number of types of cancers studied and lack of sufficient attention to lung, colorectal, and prostate cancers; (3) frequent use of multi-state SEER data; and (4) failure to empirically test potential mediators of the segregation-cancer relationship.

The studies also exhibited these eight additional, important limitations: (1) In some studies, the size of the AA sample was not specified. (2) In some studies, the area-level (e.g., census tract, county) at which segregation was measured was unclear, and in most studies, the level of segregation regarded as low, moderate, and high was never specified. (3) Some studies measured segregation at the county or zip code level even though these levels generally are too large to detect geographic area effects [10]. (4) Gender-specific analyses were not conducted for lung and colorectal cancers; hence, possible socially or biologically mediated gender differences in the segregation-cancer relationship remain unknown. (5) Possible rural-suburban-urban differences rarely were examined. Such differences were tested in only one cancer-registry study [60], and many of the cancer-risk studies focused on urban areas only. Urban/non-urban residence however contributes significantly to cancer and to cancer disparities,

among AAs in particular [1, 3, 60, 61]. Data on urban vs. suburban vs. rural residence are included in almost all cancer registries but remain unexamined as potential mediators of the segregation effect—or lack thereof. Moreover, (6) although the quality of cancer-registry data is continually monitored and there are ongoing efforts to improve it, missing data (e.g., on patient demographics and follow-up) remain a problem. Specifically, missing data on patient race is not uncommon in medical records and cancer registries. Such missing data are a limitation and potential source of bias in studies of segregation and racial cancer disparities that utilize cancer-registry data [2].

In addition, (7) patient race when not missing also is a serious concern in cancer registries. Perceived or socially assigned race (how others—healthcare providers in particular—categorize us) may or may not match self-identified race (how we categorize ourselves). Consequently, the socially assigned race often used in medical records and subsequent cancer registries might not match self-identified race [2, 79, 80, 81]. Indeed, one study found that 4% of self-identified AAs, 27% of self-identified Latinos, 48% of self-identified American Indians, and 60% of self-identified multi-racial adults were misclassified as white by others [81]. Misclassification of patient race in cancer registries is a limitation and potential source of bias in studies of segregation and cancer disparities that are based on cancer-registry data. Finally, (8) all 17 studies used census data on AAs and whites to calculate segregation, irrespective of the segregation measure used. A major limitation and bias of census data is that AAs are always under-counted because of their high (urban) residential mobility; residence in neighborhoods deemed dangerous and hence avoided by census takers; residence in remote, rural areas that lack addresses; refusal rates, and other factors [82]. The statistical adjustments commonly used to compensate for the persistent, pervasive under-counting of AAs are themselves problematic, and hence raise additional questions about census data validity [82]. Thus, the accuracy/validity of segregation data is an additional limitation of studies of segregation and cancer.

**Conclusions** Despite the aforementioned 12 limitations, and despite the differences among the studies in segregation measure and cancer registry utilized, all 17 studies were secondary data analyses of a large, existing dataset that is generally regarded as reliable and valid. Segregation was found to contribute significantly to cancer and to racial cancer disparities in 70% of analyses of such datasets. This suggests that the topic is worthy of further research. As detailed above, research is needed that uses valid measures of segregation; uses the variety of valid measures that have not been examined; assesses segregation at the CT level instead at higher levels (zip codes, counties); examines colorectal, prostate, and lung cancers; and examines the health-behavior and neighborhood variables (e.g., resources, hazards, rurality) that may mediate the segregation effect.

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